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Smoothing and Compressing Algorithm of Toolpath with Complex Contour in NC Machining

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Abstract: The continuous short line segment processing method in CNC machining has the defect of frequent fluctuation of feedrate, which affects the quality and efficiency. For this problem, the local and global methods are common solutions. The local method is to insert transition curves at the corners to obtain a smooth toolpath, but this method does not essentially reduce the number of acceleration and deceleration, and it is difficult to insert transition curves between tiny straight sections. Although the global method can obtain smooth tool trajectories, the parametric interpolation method is not suitable for low- and mid-range CNC machines. The article proposes an adaptive tool path generation method with the original part contour as the research object and discrete the NURBS curves into smooth trajectories consisting of large segments of straight lines and circular arcs. The article mimics the crawling characteristics of snakes and designs a double-headed snake algorithm based on the least squares method to implement this process. First, given the start and end points of the curve, two snakeheads search for the maximum linear and circular segments that satisfy the error constraint in the curve direction. The winner will then be selected as the candidate track segment through a competition mechanism. Finally, all trajectory segments are connected to obtain a smooth tool path. Experiments show that the toolpath data of the method in this paper is reduced by 75.56% compared with commercial CAM software when the contour error threshold is the same, and the contour error is reduced by more than one order of magnitude when the number of toolpath segments is equal. In addition, the method in this paper can obtain a smoother machining surface and more stable cutting force, thus achieving a win-win situation of quality and efficiency.

Key words: NURBS curves; Least squares method; Generation of toolpaths; Toolpaths smoothing

1. Introduction

The rapid development of computer-aided geometric design (CAGD) has enabled the precise mathematical representation of complex contours, making the machining of irregularly contoured parts a hot topic in academic research and industrial applications [1]. NURBS curves are defined by the International Organization for Standardization as the only data exchange standard for computer-aided design/manufacturing (CAD/CAM) because of their superior shape representation capabilities [2], and most commercial CAD/CAM software uses NURBS as the mathematical representation of parts [3-4]. Therefore, enhancing the processing capability of CNC machine tools to NURBS curves can improve the machining quality and efficiency of parts, and further increase the machining performance and level of machine tools.

General CNC machine tools only have linear and circular interpolation functions, which cannot accurately track the intended curve trajectory, so the curve needs to be discretized to obtain the cutting position (CL) [5-6]. Therefore, the machining of complex parts on these machines is divided into two main steps. The first is tool trajectory planning, that is, the curve is discrete into a continuous short line segment trajectory. The second is feed

rate planning, i.e., controlling the tool to machine at different speeds in different curvature regions. Among them, the generation of toolpaths has a significant impact on the quality and efficiency of the machining of the parts [7]. The discretization of parametric curves can be summarized in the following three methods, i.e., equal parameter discretization, equal chord length or equal arc length discretization, and equal chord error discretization [8]. For example, Sun et al. [9], He et al. [10], and Zou et al. [11] uniformly divide the parameters of the spline curve to obtain the toolpath. However, this method is not easy to control the contour error between the toolpath and the original curves, and it may need to adjust the division steps of the parameters several times to get the toolpath that meets the error requirement. Liu et al. [12] address the path planning problem of welding robots by discretizing the spatial curves by the equal chord difference and equal arc length methods. Du et al. [13] discretized the curves by the equal chord error method. Zhao et al. [14] searched for the maximum chord error by the golden mean method.

The discretization of the curve can be established as a polygon approximation problem, i.e., given N points of the original curve, find n characteristic points among them so that the trajectory of the continuous short line segment composed of them can replace the original curve within the error range[15]. The aim is to carry the geometric information of the original curve with as few points as possible [16-17]. Douglas- Peucker algorithm [18] (D-P) is a classic algorithm in the field of trajectory data compression. For instance, Etienne et al. [19] used the D-P algorithm to compress the location data of traffic trajectories while preserving the important locations. Zhao [20 also uses the D-P algorithm for ship trajectory compression. But the threshold value of the D-P method is difficult to determine and tends to filter the geometric information of the original trajectory.

Although the above-mentioned methods can obtain tool trajectories that meet the geometric accuracy requirements, the following two drawbacks cannot be ignored. First, the large number of short line segments means that the machine needs frequent acceleration and deceleration, which is especially obvious in high-speed machining, so the machining efficiency is greatly reduced, and the frequent vibration will also reduce the life of the machine. Secondly, the unsmooth toolpath causes the poor surface quality of the part. Nam et al. [21] proposed S-shaped feedrate planning method to improve machining efficiency. Shingo et al. [22] considered the dynamic performance of the machine tool and constructed a circular velocity transition model at the corners to improve the tool articulation speed between adjacent micro-line segments. Shi et al. [23] use a look-ahead algorithm to determine in advance whether one should accelerate or decelerate based on the curvature change of the curve. However, the above work does not essentially reduce the number of acceleration and deceleration. The research in tool trajectory smoothing can be summarized as local and global methods [24-25]. For instance, Zhang et al. [26] inserted a micro-arc at the corner to obtain a smooth tool path. Park H [27] divides the NURBS curve into discrete points, and then fits the curve with a bicircular arc between adjacent points. Fan et al. [28] and Zhang et al. [29] used quadratic Bézier curves to smooth the corners. Vincent et al. [30] and Zhao et al. [31] used cubic B-samples as transition curves. Zhang et al. [32] inserted five NURBS curves at the corners of the continuous linear tool path to obtain the G4 continuous toolpath.

Although the local method can obtain smooth tool trajectories, the length of many trajectory segments in actual machining is in the micron range, which will increase the difficulty of the algorithm or even make it ineffective [33-34]. In addition, there is a large amount of research focusing on global approaches, i.e., reconstructing short line segment trajectories as parametric splines. For example, Zhang et al. [35] and Yau et al. [36] first generated the G-code data by CAM software and then fitted it to a NURBS curve. Liu et al. [37] compressed the huge amount of tool location data by double-chord error and then reconstructed it into NURBS curves by the least squares method. References [38-41] considered from the perspective of reducing the complexity of the algorithm by first selecting a set of dominant points on the original data points and then fitting them to the B-spline. Although the global method can obtain a smooth toolpath, unfortunately only a few CNC machine tools with parametric curve interpolation function, so it is not easy to promote in ordinary machine tools [42]. Almost all CNC machines support linear and circular interpolation, so the G-code command interpolation

method is still dominant in the industry [43].

To achieve a higher level of quality and efficiency for ordinary CNC machines, this paper considers both the compression of the number of trajectory segments and trajectory smoothing, and directly discrete NURBS into a combination of large-sized linear and circular arc segments. In this paper, a double-headed snake algorithm (DHS) based on least squares is designed to implement the process. The two snake heads represent the linear segment search engine (LSSE) and the arc segment search engine (ASSE), respectively. LSSE searches for the longest linear segment that meets the accuracy requirement by the forward-backward search strategy, and ASSE searches for the largest arc segment that meets the accuracy requirement by the least-squares method, and then determines the next trajectory segment by the campaign mechanism, and finally checks whether the current trajectory segment satisfies the Smoothness condition. The content of the article is organized as follows: Section 2 is the mathematical foundation, mainly including the definition and properties of NURBS curves and the least squares-based arc fitting method. Section 3 is the core of the approach in this paper and will describe the details of the two-headed algorithm. Section 4 will categorically discuss the smoothing conditions between straight and circular segments and derive the corresponding formulas. Section 5 is the simulation and experimental analysis, and Section 6 is the conclusion.

2. Foundations of mathematical knowledge

2.1 The definition and derivation of NURBS

A k times NURBS curve can be defined as

$$C(u)=\frac{\sum_{i=0}^{n}N_{i,k}(u)\omega_i P_i}{\sum_{i=0}^{n}N_{i,k}(u)\omega_i} \qquad u\in[0,1] \tag{1}$$

Where $\{P_i\}$ is the control vertices that form the control polygon, $\{\omega_i\}$ is the weight corresponding to each control vertex, and $\{N_{i,k}\}$ is the n+k+2nd kth B-spline basis function defined by the non-uniform node vector U, and

$$U=\{\underbrace{0,...,0}_{k+1},u_{k+1},u_{k+2},...,u_n,\underbrace{1,...,1}_{k+1}\} \tag{2}$$

$$\begin{cases} N_{i,0}=\begin{cases}1, & u_i\le u<u_{i+1}\\ 0, & else\end{cases} \\ N_{i,k}=\dfrac{u-u_i}{u_{i+k}-u_i}N_{i,k-1}+\dfrac{u_{i+k+1}-u}{u_{i+k+1}-u_{i+1}}N_{i+1,k-1}\end{cases} \tag{3}$$

Eq. (3) is called the de Boor-Cox recurrence formula. In particular, Eq. (1) degenerates to a non-uniform B-sample curve when the weight factor ω_i=1. If the vector values in Eq. (2) are uniformly varying except for k + 1 nodes at the ends, the curve further degenerates to a quasi-uniform B-spline. In order to obtain the exact curvature of the curve, the derivative of the NURBS curve is required. First, the derivatives of the B-spline and basis functions are given without proof as follows.

$$N_{i,k}^{(r)}=k\left(\frac{N_{i,k-1}^{(r-1)}}{u_{i+k}-u_i}-\frac{N_{i+1,k-1}^{(r-1)}}{u_{i+k+1}-u_{i+1}}\right) \tag{4}$$

$$\begin{cases} C_{i,k}^{(r)} = \sum_{i=0}^{n-r} N_{i,k-r} P_i^{(k)} \\ P_i^{(r)} = \begin{cases} P_i \ , & k=0 \\ (k-r+1)\dfrac{P_{i+1}^{(r-1)} - P_i^{(r-1)}}{u_{i+k+1} - u_{i+k}} & k \neq 0 \end{cases} \end{cases} \tag{5}$$

Then according to reference [4], let

$$C(u) = \frac{\omega(u)C(u)}{\omega(u)} = \frac{A(u)}{\omega(u)} \tag{6}$$

Where $A(u) = \sum_{i=0}^{n} N_{i,k} P_i \omega_i$ is a vector-valued function and is the numerator of Eq. (1), $\omega(u) = \sum_{i=0}^{n} N_{i,k} \omega_i$. The r-order derivative of $A(u)$ is obtained by Leibniz's formula as

$$A^{(r)}(u) = [\omega(u)A(u)]^{(r)} = \sum_{i=0}^{r} \binom{r}{i} \omega^{(i)}(u) C^{(r-i)}(u) = \omega(u)C^{(r)}(u) + \sum_{i=1}^{r} \binom{r}{i} \omega^{(i)}(u) C^{(r-i)}(u)$$

Thus the formula for the r-order derivative of the NURBS curve can be obtained as

$$C^{(r)}(u) = \frac{A^{(r)}(u) - \sum_{i=1}^{r} \binom{r}{i} \omega^{(i)}(u) C^{(r-i)}(u)}{\omega(u)} \tag{11}$$

Where $A^{(r)}$, $C^{(r)}(u)$ and $\omega^{(i)}(u)$ can be obtained from Eq. (5).

2.2 The selection of feature points

Arcs are curves of equal curvature, and when using arc segments to approximate complex curves, each arc segment usually does not include multiple extreme points or inflection points of the curve. Therefore, the interval where the end point of the circular arc segment is located can be determined by the inflection point and the extreme value point, thus speeding up the convergence of the algorithm. Usually, the curvature is positive, which only reflects the degree of bending of the curve, while the relative curvature can be positive or negative, which reflects both the degree and direction of bending of the curve. The relative curvature is calculated as follows:

$$\kappa(u_i) = \frac{\det[C^{(1)}(u_i) \quad C^{(2)}(u_i)]}{\left|C^{(1)}(u_i)\right|^3} \tag{12}$$

A negative value means that the curve is deflected clockwise along the parameter direction, while a positive value means counterclockwise. If the curvature value changes sign, it means that the point is the inflection point of the curve, as shown in Fig. 1.

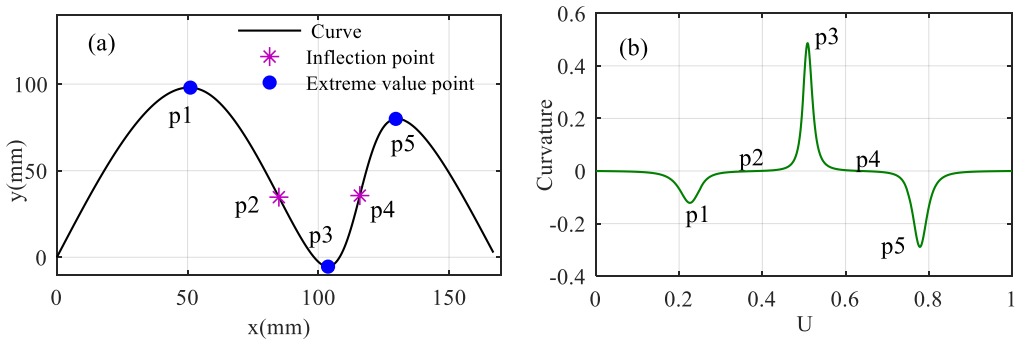

Fig. 1 Obtain feature points by relative curvature. (a) Curves and feature points.(b) Relative curvature

The extreme points in Fig. 1(a) correspond to the extreme points of curvature in Fig. 1(b), and the inflection point is the point where the relative curvature is zero and the left and right sides have different signs. The extreme value point and the inflection point of the curve should satisfy the following conditions, respectively.

$$P_{extre} = \{P(u_i) \left| \kappa(u_{i-1}) < \kappa(u_i) \cap \kappa(u_i) > \kappa(u_{i+1}), P(u_i) \right| \kappa(u_{i-1}) > \kappa(u_i) \cap \kappa(u_i) < \kappa(u_{i+1})\}$$

$$P_{inflec} = \{P(u_i) \left| \kappa(u_{i-1}) \cdot \kappa(u_{i+1}) < 0 \right.\}$$

Therefore, adding the starting and ending points of the curve, the set of characteristic points of the curve is

$$P_f = \{P_{start}, P_{inflec}, P_{extre}, P_{end}\}$$

2.3 Arc fitting based on least squares method

The least squares method is a simple and effective optimization method, and the idea is to find the unknown parameters in the objective function by minimizing the sum of squared errors given a set of samples.

Given a sequence P of points on a complex curve, suppose P_s is the starting point, P_e is the ending point, and the arc segment passes through P_s and P_e, then the least squares-based arc fitting objective function can be expressed as

$$F=\sum\left[(x_i-P_{mx}-ke_1)^2+(y_i-P_{my}-ke_2)^2-(P_{sx}-P_{mx}-ke_1)^2+(P_{sy}-P_{my}-ke_2)^2\right]^2 \tag{13}$$

Where P_m is the midpoint of P_sP_e, and $\boldsymbol{e}=(e_1,\ e_2)$ is the normal unit vector of P_sP_e. Let $x_i-P_{mx}=\bar{x}_i$ $y_i-P_{my}=\bar{y}_i$, then the objective function can be reduced to

$$F=\sum\left[\bar{x}_i^2+\bar{y}_i^2+(B-2e_1\bar{x}_i-2e_2\bar{y}_i)k-A\right]^2 \tag{14}$$

Where $A=(P_{sx}-P_{mx})^2+(P_{sy}-P_{my})^2$, $B=2e_1(P_{sx}-P_{mx})+2e_2(P_{sy}-P_{my})$. According to the extreme value condition, let the first order derivative of Eq. (14) be zero to find the unknown parameter as

$$k=(AH_3-H_1-H_2)/H_3 \tag{15}$$

Where $H_1=\sum\left[\bar{x}_i^2(B-2e_1\bar{x}_i-2e_2\bar{y}_i)\right]$, $H_2=\sum\left[\bar{y}_i^2(B-2e_1\bar{x}_i-2e_2\bar{y}_i)\right]$, $H_3=\sum(B-2e_1\bar{x}_i-2e_2\bar{y}_i)^2$ .Therefore, the equations for the center and radius of the circle can be obtained as follows

$$\begin{cases} O=P_m+ke \\ r=\sqrt{(P_{mx}-O_x)^2+(P_{my}-O_y)^2} \end{cases} \tag{16}$$

It should be noted that the arc can only be obtained by Eq. (16) in the least squares sense, and the arc segment may not meet the actual machining accuracy requirements. Assuming that the threshold value of the chord error is δ_{con}, the following constraints should be added:

$$\max\left(\left|\sqrt{(x_i-O_x)^2+(y_i-O_y)^2}-r\right|\right)\le\delta_{con}$$

3. The generation of toolpaths

3.1 The search for the largest arc segment

From mathematical knowledge, it is easy to define that a circle can be uniquely determined by any three points in the plane that are not co-linear, and that the radius of this circle tends to infinity when the three points are co-linear. Given a certain starting point P_{start} on the curve, in order to obtain the maximum arc that satisfies the contour accuracy, the following two corollaries are given.

Corollary 1: For an arbitrary set of discrete columns of points captured on a curve, we can always find a unique arc to fit these points in a least squares sense and minimize the contour error.

Corollary 2: Pick an arbitrary starting point P_{start} on the curve, increase the search point P_{sch} from that point and perform the arc fitting operation. When the column of points between P_{start} and P_{sch} no longer meet the fitting accuracy requirements, continue to increase the search points also do not meet the requirements, while the points between P_{start} and P_{sch} are satisfied with the fitting conditions.

Since each set of points can be uniquely solved by the method in Section 2.3, the correctness of corollary 1 is obvious. Unfortunately, corollary 2 is not easy to prove mathematically, but this conclusion was drawn from our experiments on partial curves, so it is not a bad idea to explore a new approach from the point of view of practical applications. Therefore, based on the above inference, it is feasible to search for the maximum arc segment

meeting the accuracy requirement along the curve direction. The article searches for the maximum arc segment that meets the accuracy requirement by the interval dichotomy method shown in Fig. 2.

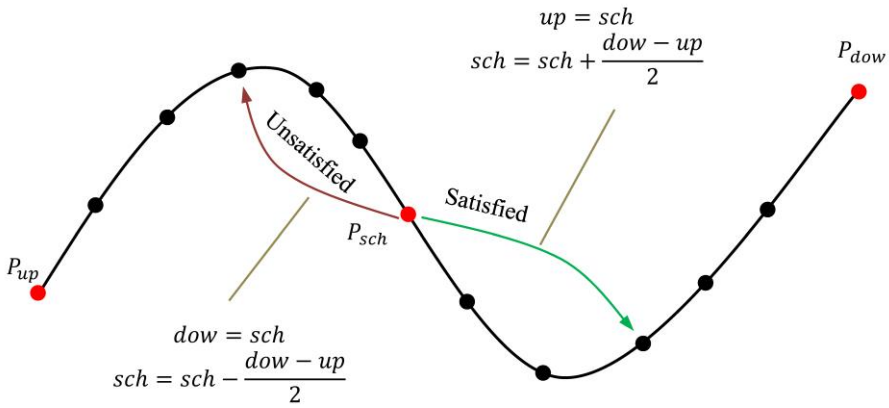

Fig. 2 Search for maximum arc by interval dichotomy

Let P be the point cloud data obtained from the curve, P_{start} and P_{end} be the start and end points of the curve, respectively. The search point P_{sch} is the moving point between P_{start} and P_{end}, and when the algorithm converges, P_{start} and P_{sch} are the starting and ending points of the arc segment and at this point, P_{sch} is called the target point. Suppose the target point is located in the interval $[P_{up}, P_{dow}]$ See in Fig. 2, then the maximum arc can be confirmed by Algorithm 1

Algorithm 1: Search for the largest arc segment

Input: point cloud P, feature point set P_f
Output: Start, end, center and radius of the largest arc segment
Step1: Initialization interval
Step1.1: Let $[P_{up}, P_{dow}] = [P_{start}, P_{end}]$, given $P_{sch} \in P_f$ and satisfying P_{sch} is the first feature point after P_{start}.
Step1.2: Perform the arc fitting operation in section 1.2 on the interval $[P_{start}, P_{sch}]$, if the accuracy meets the requirement, then let $P_{up} = P_{sch}$, $sch = sch + 1$, continue the above operation, otherwise make $P_{dow} = P_{sch}$, end step1.
Step2: Narrowing the interval
Step2.1: Let $P_{sch} = P_{up}$ and $P_{sch} \in P$.
Step2.2 Perform the arc fitting operation in section 1.2 on the interval $[P_{start}, P_{sch}]$
Step2.3: If the fitting result meets the accuracy requirement, make $P_{up} = P_{sch}$, $t = 0.5(dow - up)$, $sch = sch + t$, otherwise make $P_{dow} = P_{sch}$, $t = 0.5(dow - up)$, $sch = sch - t$
Step2.4 if $dow - up > 1$, return to step2.2, otherwise end Step2.
Step3: Return calculation results
Perform the arc fitting operation on $[P_{start}, P_{up}]$, output the corresponding center and radius, and the starting and ending points of the arc are P_{start} and P_{up}

3.2 The search for the longest straight line segment

The core idea of this part is to make the search point P_{sch} move along the curve direction from the starting point P_{start} and find the longest straight line segment $P_{start}P_{sch}$, so that the contour error between the line segment $P_{start}P_{sch}$ and the curve meets the requirement. In order to speed up the algorithm, this paper adopts the forward-back search strategy with variable step size, and the specific steps are shown in Algorithm 2

Algorithm2: Search for the longest linear segment

Input: point cloud P
Output: The start and end of a straight line segment
Step1: Given an initial search point $P_{sch} = P_{start}$ and $P_{sch} \in P$, initialization step t=1
Step2: Start Searching
Step2.1: Let $sch = sch + t$
Step2.2: Calculate the chord error E between the line $P_{start}P_{sch}$ and the curve

Step2.3: if E< δ_{con} , let t=2t, return Step2.1. Else if E= δ_{con} , go to Step3. Else if E> δ_{con} , if t=1, let $sch = sch - 1$, go to Step3, otherwise, let $sch = sch - t$, t=1, return Step2.1.

Step3: Output linear segment $P_{start}P_{sch}$.

3.3 Overall design

The bodies of snakes have excellent sinuous nature and they can crawl along arbitrarily complex trajectories. In this paper, a double-headed snake search algorithm is designed to implement the toolpath generation process by this property. The main structure of the algorithm is shown in Fig. 3.

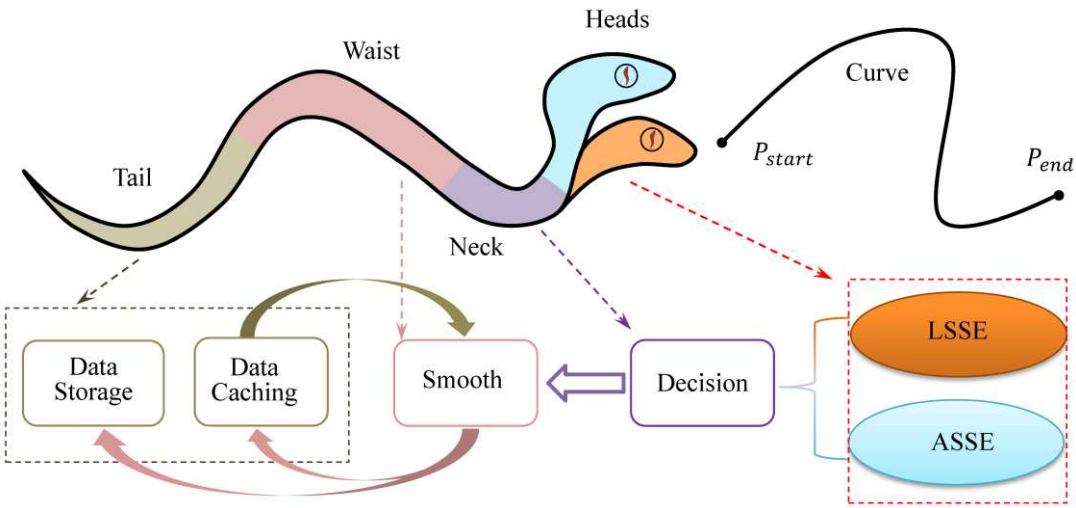

Fig. 3 Composition of the double-headed snake algorithm

The DHS algorithm consists of four parts: the heads, the neck, the waist, and the tail. The head includes two independent search engines, The LSSE searches for the longest straight segment (seg_l) by Algorithm 2 and ASSE searches for the largest arc segment (seg_a) by Algorithm 1 when the snake is crawling along the curve from the starting point. The neck sets up a campaign mechanism to select the best trajectory segment (seg_best) from seg_l and seg_a .For the best data compression, the algorithm selects the segment with the largest span in the curve direction as the winner. The waist is the smoothing module of the algorithm, whose function is to check whether the smoothing conditions are satisfied between the current trajectory segment $seg_best(i)$ and the previous trajectory segment $seg_best(i-1)$ .The tail is the data storage module, which mainly includes data cache and data storage. The data cache records only the last trajectory segment and is used for smoothing condition checking of the trajectory. Data storage is used to record all trajectory segments until the end of the algorithm.

The P_{start} is pushed forward at the end of each search round, so that the curve is discretized into a combination of straight and circular segments as the snake climbs from the start to the end of the curve.

4. Conditions for trajectory smoothing

There are three cases of connection between adjacent trajectory segments, straight line to straight line, straight line to circular arc segment and arc segment to arc segment.

1) Straight line to straight line

Fig. 4(a) shows the smoothing model of a straight line transferring to a straight line, in which P_{start} is the starting point of the current straight line segment and also the ending point of the previous trajectory segment, $\boldsymbol{v}$ and $\boldsymbol{u}$ are the unit vectors of these two trajectory segments along the forward direction, respectively. To ensure the quality of machining, when the angle between $\boldsymbol{v}$ and $\boldsymbol{u}$ exceeds a given value, it is necessary to insert a transition arc between these two trajectory segments.

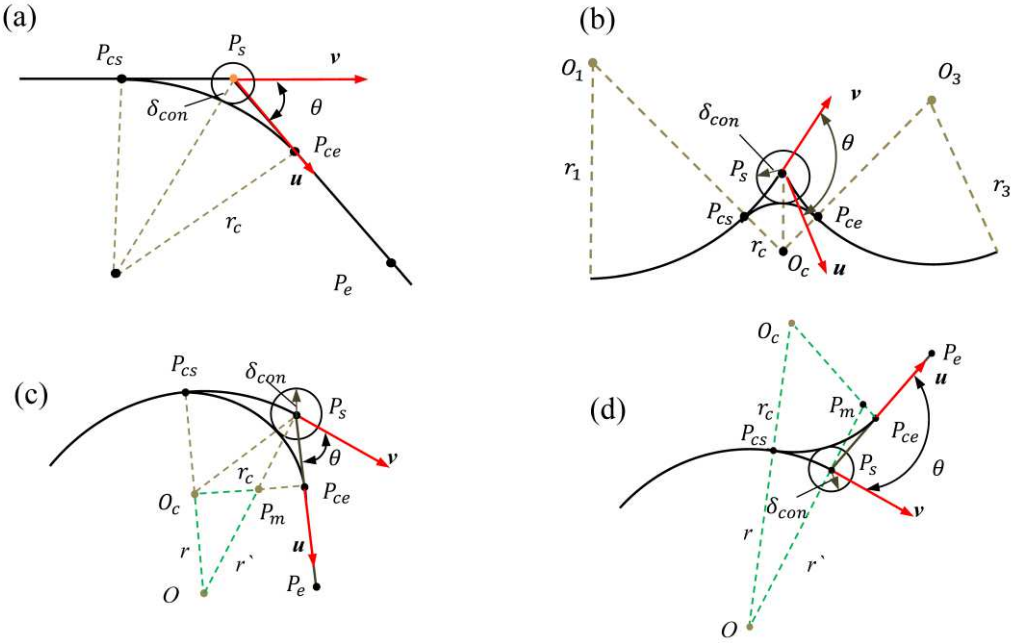

Fig.4 Smoothing of toolpath. (a) Straight line to straight line, (b) Arc segment to arc segment, (c) Straight line to arc(Same direction arc), (d) Straight line to arc(Reverse direction arc).

In the fig.4 (a), P_{cs} and P_{ce} are the starting and ending points of the transition arc, O_c and r_c are the center and radius, respectively. The circle with P_{cs} as the center and the error constraint δ_{con} as the radius is the error circle, so the maximum contour error occurs at P_{cs}. From the geometric relationship, we have

$$r_c = \delta_{con}\cos(\theta/2)/(1-\cos(\theta/2)) \tag{17}$$

The starting and ending coordinates of the transition arc are

$$\begin{cases} P_{cs} = P_s - r_c\tan(\theta/2)\cdot \boldsymbol{v} \\ P_{ce} = P_s + r_c\tan(\theta/2)\cdot \boldsymbol{u} \end{cases} \tag{18}$$

2) Arc segment to arc segment

As shown in Fig. 4 (b), O_1 and O_3 are adjacent arc segments, then the transition arc O_c should be tangent to these two arc segments and the error circle, according to the geometric relationship is not difficult to derive

$$\begin{cases} (O_c(x)-O_1(x))^2+(O_c(y)-O_1(y))^2=(r_c+r_1)^2 & ① \\ (O_c(x)-P_s(x))^2+(O_c(y)-P_s(y))^2=(r_c+\delta_{con})^2 & ② \\ (O_c(x)-O_3(x))^2+(O_c(y)-O_3(y))^2=(r_c+r_3)^2 & ③ \end{cases} \tag{19}$$

Let $a_1 = P_s(x)-O_1(x), b_1 = P_s(y)-O_1(y), c_1 = r_1-\delta_{con}, a_2 = O_3(x)-P_s(x), b_2 = O_3(y)-P_s(y), c_2 = \delta_{con}-r_3$ $d_1 = (r_1^2-\delta_{con}^2+P_c^2(x)-O_1^2(x)+P_c^2(y)-O_1^2(y))/2$, $d_2 = (\delta_{con}^2-r_3^2+O_3^2(x)-P_c^2(x)+O_3^2(y)-P_c^2(y))/2$. Then, the Eq. (19) can be simplified by combining ①-② and ②-③ as

$$\begin{cases} O_c(x) = e_1 r_c + f_1 \\ O_c(y) = e_2 r_c + f_2 \end{cases} \tag{20}$$

Where $e_1 = (c_1b_2-b_1c_2)/(a_1b_2-b_1a_2)$, $e_2 = (c_1a_2-a_1c_2)/(b_1a_2-a_1b_2)$, $f_1 = (d_1b_2-b_1d_2)/(a_1b_2-b_1a_2)$ $f_2 = (d_1a_2-a_1d_2)/(b_1a_2-a_1b_2)$. Substituting Eq. (20) into ① yields the following quadratic equation for r

$$h_1 r_c^2 + h_2 r_c + h_3 = 0 \tag{21}$$

Where $h_1 = (e_1^2+e_2^2-1)$, $h_2 = 2e_1(f_1-O_1(x))+2e_2(f_2-O_1(y))-2r_1$, $h_3 = (f_1-O_1(x))^2+(f_2-O_1(y))^2-r_1^2$. The radius of the transition arc can be obtained by Eq. (21), and it should be noted that the equation has two solutions, positive and negative, which correspond to the cases of externally and internally tangent, respectively. Since the transition arc is tangent to the three circles, the result of the calculation should take a positive value.

3) Straight line to arc

Figs 4(c) and (d) show the transition in the same direction and in the opposite direction, respectively. In this

figure, $\boldsymbol{O}$ and $\boldsymbol{r}$ are the center and radius of seg_a, respectively, then the transition arc should be tangent to seg_a, the error circle and seg_l at the same time. Let OP_s and O_cP_{ce} intersect at P_m, $|OP_m| = r`$, then there is the following equation for the geometric relationship in Fig. 4(c).

$$\begin{cases} (r-r`)\sin(\theta) - \sqrt{2r_c\delta_{con} + \delta_{con}^2} = 0 \\ \cos(\theta) - \dfrac{r`+[r_c-(r-r`)\cos(\theta)]^2 - (r-r_c)^2}{2r`[r_c-(r-r`)\cos(\theta)]} = 0 \end{cases} \tag{22}$$

Let $a = \sin^2(\theta) + \delta_{con}/r$, $b = -2r\sin^2(\theta)$, $c = r^2\sin^2(\theta) - r\delta_{con}(1+\cos(\theta)) - \delta_{con}^2$, then Eq. (22) can be reduced to

$$ar`^2 + br` + c = 0 \tag{23}$$

Since the inserted arc is always inside the circle $\boldsymbol{O}$, and the radius is positive, the solution of Eq. (23) should be the smaller of the absolute values.

$$r` = \min\left(\left|\frac{-b+\sqrt{b^2-4ac}}{2a}\right|, \left|\frac{-b-\sqrt{b^2-4ac}}{2a}\right|\right)$$

Substituting $r`$ into Eq. (22) yields the radius of the transition arc as

$$r_c = \frac{r^2(1+\cos(\theta)) - r`^2}{2r} \tag{24}$$

The start, end and center of the transition arc are calculated by

$$\begin{cases} P_{ce} = P_s + \sqrt{(r_c+\delta_{con})2 - \delta_{con}^2} \cdot \boldsymbol{u} \\ O_c = P_{ce} + r_c \cdot \boldsymbol{u}` \\ P_{cs} = O + r \cdot OO_c / \|O_c - O\| \end{cases} \tag{25}$$

Where $\boldsymbol{u}`$ denotes the vertical unit vector of $\boldsymbol{u}$. When the transition arc is shown in Fig. 4(d), Eq. (22) becomes

$$\begin{cases} (r`-r)\sin(\theta) - \sqrt{2r_c\delta_{con} + \delta_{con}^2} = 0 \\ \cos(\pi-\theta) - \dfrac{r`^2+[r_c-(r`-r)\cos(\theta)]^2 - (r+r_c)^2}{2r`[r_c-(r`-r)\cos(\theta)]} = 0 \end{cases} \tag{26}$$

And let $a = \sin^2(\theta) + \delta_{con}(1+\cos(\theta))/r$, $b = -2r\sin^2(\theta)$, $c = r^2\sin^2(\theta) + r\delta_{con}(1+\cos(\theta)) - \delta_{con}^2$ .Since the transition arc is externally tangent to the circle $\boldsymbol{O}$, substituting a, b, and c into Eq. (23) yields

$$r` = \max\left(\left|\frac{-b+\sqrt{b^2-4ac}}{2a}\right|, \left|\frac{-b-\sqrt{b^2-4ac}}{2a}\right|\right) \tag{27}$$

Substituting Eq. (27) into Eq. (22) yields

$$r_c = \frac{r`^2(1+\cos(\theta)) - r_1^2(1+\cos(\theta))}{2r} \tag{22}$$

The rest of the unknown quantities are calculated by the same formula as Eq. (25).

5. Simulation and experimental analysis

5.1 The generation of toolpath and error analysis

To verify the feasibility and effectiveness of the method in this paper, the "dolphin" curve with the length of 675.6525 shown in Fig. 5 was used for the experiment. This method is compared with the equal chord error method, the D-P algorithm, and the commercial CNC software Mastercam. The advantages and disadvantages of this method are analyzed in terms of contour accuracy, compression efficiency (number of discrete points of the trajectory), and program running time.

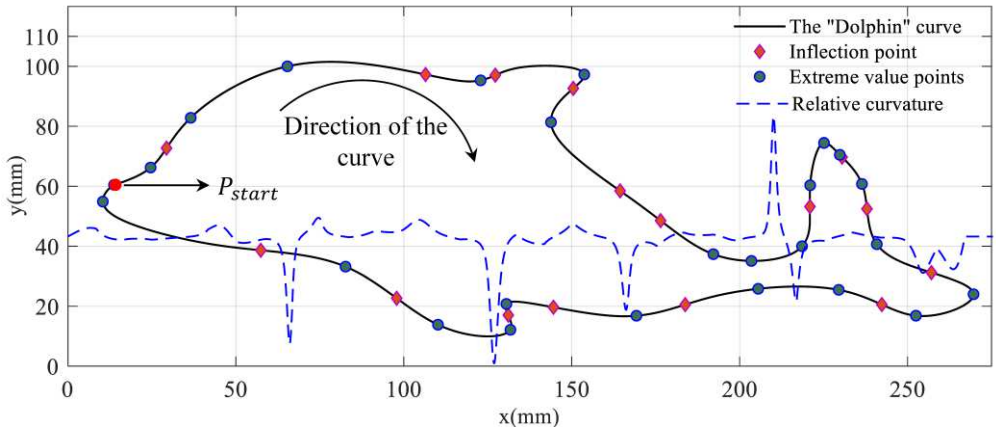

Fig.5 The "dolphin" curve and its characteristic points

The experimental platform is Dell OptiPlex 390, processor is Intel(R) Pentium CPU G630@2.7GHZ, Windows 10 operating system, software version is matlab2018a. 30030 data points are collected uniformly on the curve as the initial iteration points for the equal chord error method and the D-P method. For the fairness of the algorithm, the method in this paper partitions the parameter U at an interval of 0.0000333 to obtain 30030 points. Since the Mastercam method is generated directly by the software, it does not participate in the speed evaluation of the algorithm. Setting the contour accuracy constraint to δ_{con}=0.1, the experimental situation is as follows.

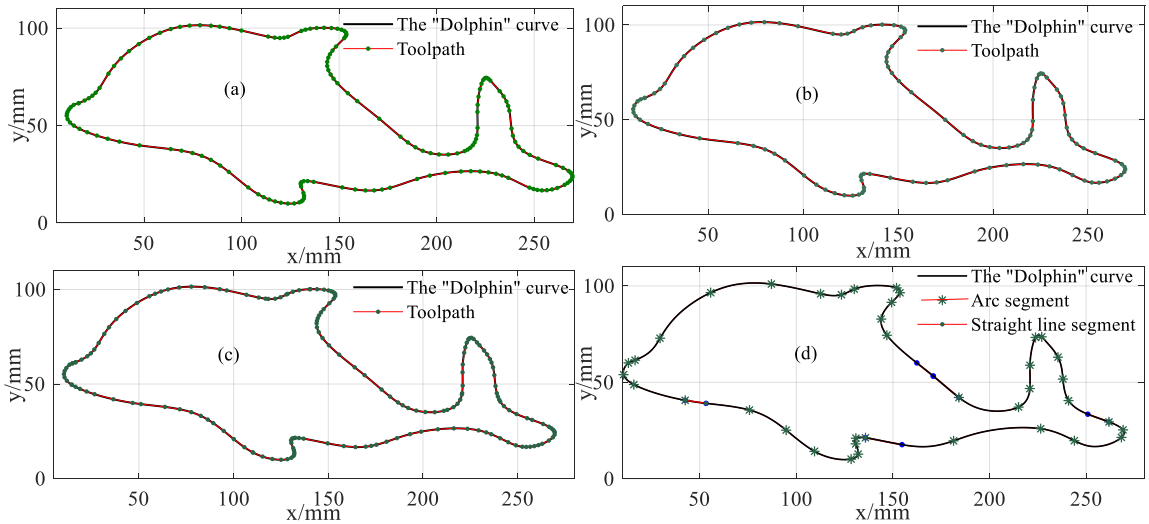

Fig.6 Toolpaths generated by the four methods. (a). D-P algorithm, (b). Equal chord error method, (c). Master-CAM, (d). The DHS algorithm

Fig. 6 shows the toolpaths generated by the four methods, from the figure it can be seen that the D-P method, the equal chord error method, and the Mastercam all approximate the original curve in the form of continuous straight line segments, and the greater the curvature the more discrete points. The DHS algorithm discrete the original curve into a combination of circular and linear segments, and the number of trajectory segments is greatly reduced. The detailed data are shown in Table 1

Table 1 Experimental data for the profile error threshold e=0.1

Methods	N	n	$\delta_{\max}$ / mm	t/s	L/mm
D-P algorithm	30030	168	0.1	103.13	675.1715
Equal chord error method	30030	122	0.1	0.78	674.1903
Mastercam	30030	180	0.1	/	675.0773
DHS algorithm	30030	43	0.1	1.54	675.5820

In Table 1, N denotes the initial iteration point, n denotes the discrete points of the tool trajectory, $\delta_{\max}$

denotes the maximum chord error, and t is the program running time. Since the memory and CPU usage of the computer is a dynamic process, the time in the table is the average value of the program run 10 times. It is clear from the data in the table1 that the toolpath generated by this method (DHS algorithm) has the least number of data points under the same accuracy constraint, which is 74.4% less than the D-P algorithm, 63.93% less than the equal chord error method and 75.56% less than the Mastercam method, respectively.

In terms of program speed, the Mastercam method was not evaluated because it was not easy to test the running time. Among the remaining three methods, the D-P method has the longest run time and the equal chord error method has the shortest run time. Although the DHS algorithm has a longer runtime than the equal chord error method, it still maintains a fast speed of 1.54 seconds. In terms of contour accuracy, all four methods met the established requirements. In addition, the difference between the total length of the trajectory and the length of the original curve can reflect the accumulated error to a certain extent. From the data in the table, it can be seen that the total length of the trajectory generated by the DHS algorithm is closer to the true value, and therefore the cumulative error is smaller.

The data in Table 2 reflects the performance of the four methods in terms of contour accuracy when the number of discrete points is the same.

Table 2 Experimental data with the same number of discrete points

Methods	N	n	E_{max} / *mm*	t/s	L/mm
D-P algorithm	30030	100	0.285	65.64	674.4373
Equal chord error method	30030	100	0.15	1.12	674.3828
Mastercam	30030	100	0.35	/	673.8129
DHS algorithm	30031	100	0.0095	1.59	675.6522

As can be seen from the data in Table 1, when n=100, the maximum contour error of the DHS algorithm is reduced by 96.67% compared to the D-P algorithm, 93.67% compared to the equal chord error, and 97.29% compared to the commercial software Mastercam. The reduction is more than one order of magnitude, which means that the method in this paper only needs to set a smaller error threshold. In addition, the difference between the trajectory length of the DHS algorithm and the original curve length is only 0.0003mm, which remains the best among the four methods.

5.2 The analysis of speed fluctuations

The feed rate of high-speed machine tools can generally reach several hundred millimeters per second or more. Due to the acceleration and deceleration capability of the machine tool itself, the machining method of continuous short line segments poses difficulties and challenges for high-speed machining due to the presence of frequent acceleration and deceleration. Therefore, the main purpose of this section is to verify whether the tool trajectory generated by DHS with the same speed control algorithm has a superior speed fluctuation performance compared to other methods. Due to the lack of high-speed machine equipment in the laboratory, the analysis of speed fluctuations was performed only by simulation.

The circular corner velocity articulation model in the literature [26] is used and tested by a seven-segment S-shaped velocity planning algorithm. Set the maximum acceleration A = 500mm/s^2, Jerk J = 5000mm/s^3, and programming speed F = 100mm/s. The experimental results are shown below:

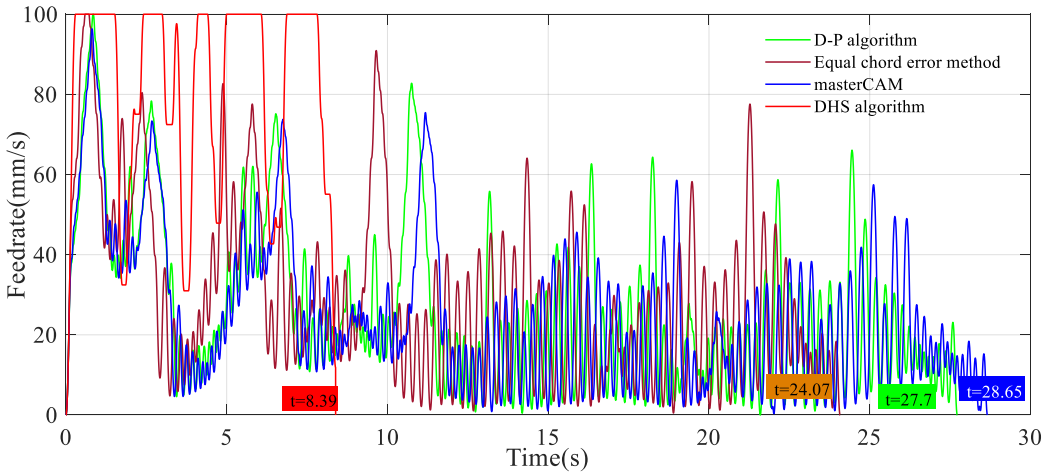

Fig. 7 The fluctuation curves of federate

Fig. 7 shows the feed speed fluctuation curve obtained by the S-type speed planning algorithm, from the figure it is obvious that the DHS algorithm can achieve a higher feedrate, so it can greatly improve the machining efficiency. The DHS algorithm reduces the machining time by 69.71%, 65.14%, and 70.71% compared to the D-P algorithm, the isospin error method and Mastercam, respectively.

The above experimental results demonstrate that the DHS algorithm can reduce the number of acceleration and deceleration by compressing the number of toolpath segments. In addition, the reduction of the number of trajectory segments means that the length of each trajectory segment increases, which gives the machine enough time to accelerate to a higher speed. Therefore, this method in this paper both slows down the speed fluctuation and improves the machining efficiency.

5.3 Analysis of physical experiments

To verify the realistic possibility of the proposed method in this paper, we experimented with a vertical CNC machine. For cost reasons, the experiment was conducted only with the commercial software Mastercam as a comparison. The equipment used in this experiment is VDL-600A vertical machining center, piezoelectric three-way force measuring instrument YDCB-III05B, and a digital microscope, as shown in Fig. 7.

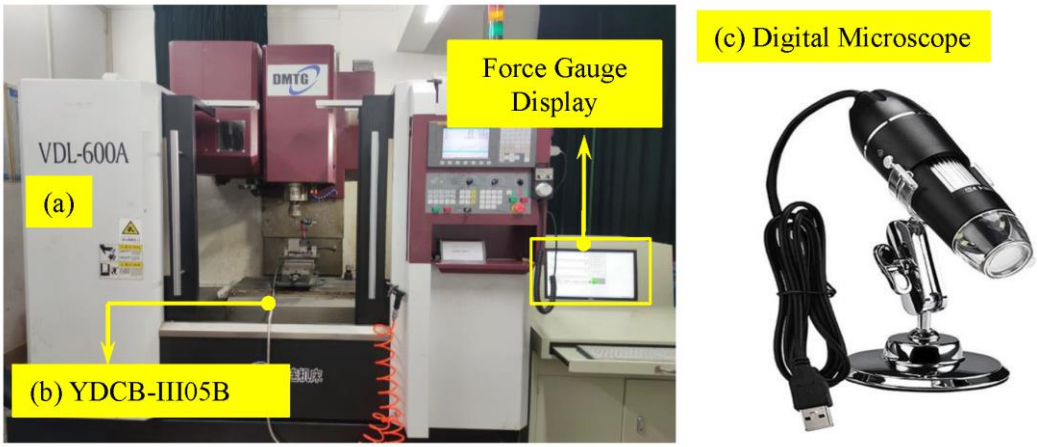

Fig. 7 Experimental equipment

The proposed method will be analyzed in terms of the smoothness of the machined surface and the fluctuation of the combined cutting force. Set the contour error threshold $\delta_{con} = 0.1$ mm, and the material is 45 carbon steel, the experimental results are shown as follows:

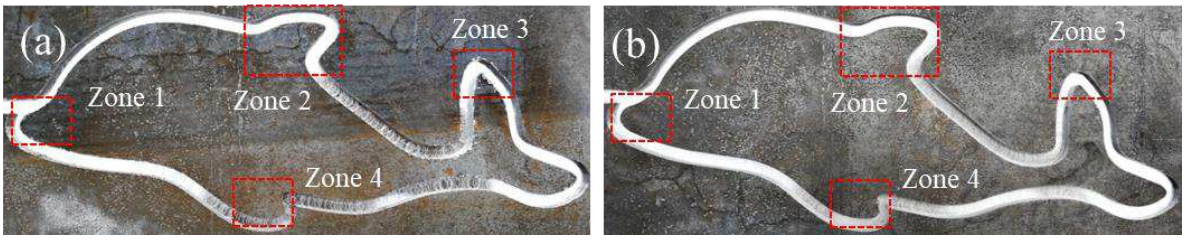

Fig. 8 The overall machining effect.(a)The Mastercam method. (b) The DHS algorithm

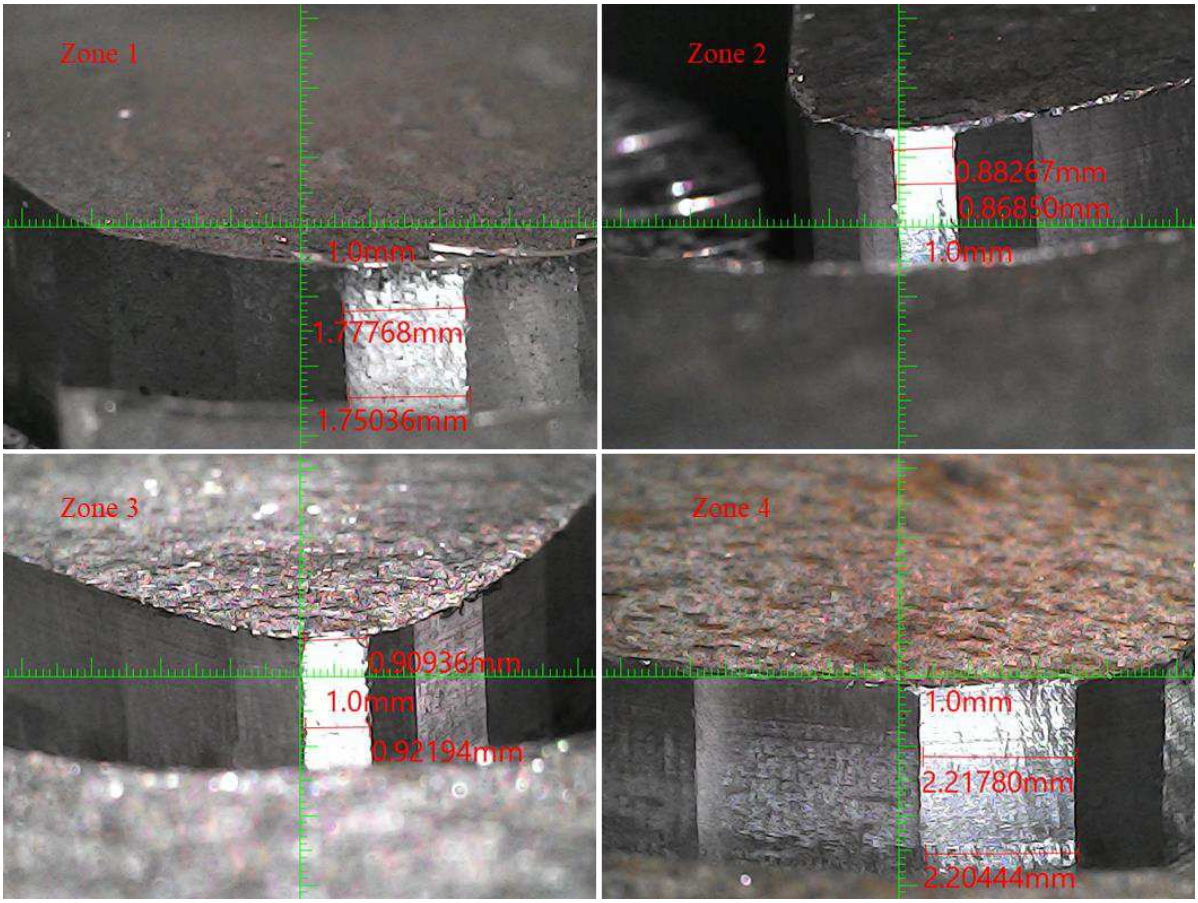

Fig. 9 Local enlargement of the masterCAM method

Fig. 8 shows the overall machining results of the Mastercam method and the DHS algorithm proposed in this paper. The red dashed boxes in the figure mark the four regions of large curvature on the curve, and the sides of the above regions were photographed by digital microscopy, and the results are shown in Fig. 9 and Fig. 10. Fig. 9 shows the local enlargement of the 4 large curvature regions of the Mastercam method. The green line in the figure is the measurement scale of the microscope, the length of each scale is 0.1mm, and the red numbers are the length data obtained by the measurement function.

It is clear from Fig. 9 that the surface of the part is not smooth in these large curvature regions, even though the profile error threshold satisfies the set conditions. The measurements of the microscope show that the minimum length of the cut segment in the local area is about 0.86 mm, while the maximum exceeds 2 mm. This means that obtaining a smoother machined surface will require a huge amount of discrete points. In contrast, Fig. 10 shows the local zoomed-in view of the DHS algorithm in the above mentioned region. From the figure, it is clear that under the same error threshold constraint, the method in this paper can obtain better surface quality while maintaining a smaller number of trajectory segments. This means that the method in this paper requires only a small number of toolpath segments to achieve superior machining quality while maintaining contour accuracy.

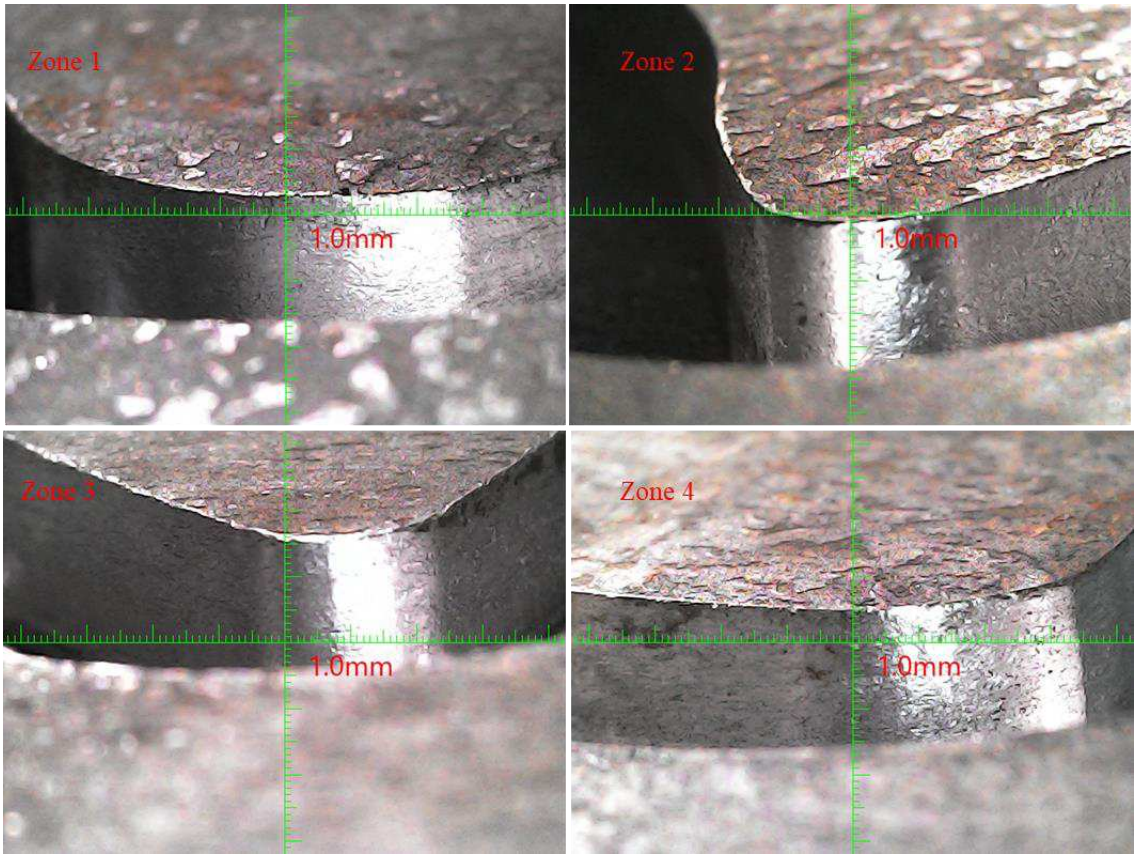

Fig. 10 Local enlargement of the DHS algorithm

Cutting force is a critical factor that is often considered in CNC machining. Frequent changes in cutting forces can cause shock and vibration in the machine tool, which can reduce tool and equipment life and even affect machining quality. For the fluctuation of cutting force, we only intercepted the force data of zone 2 for analysis because the force data generated during the whole machining process was too large and not conducive to analysis. As shown in Fig. 11.

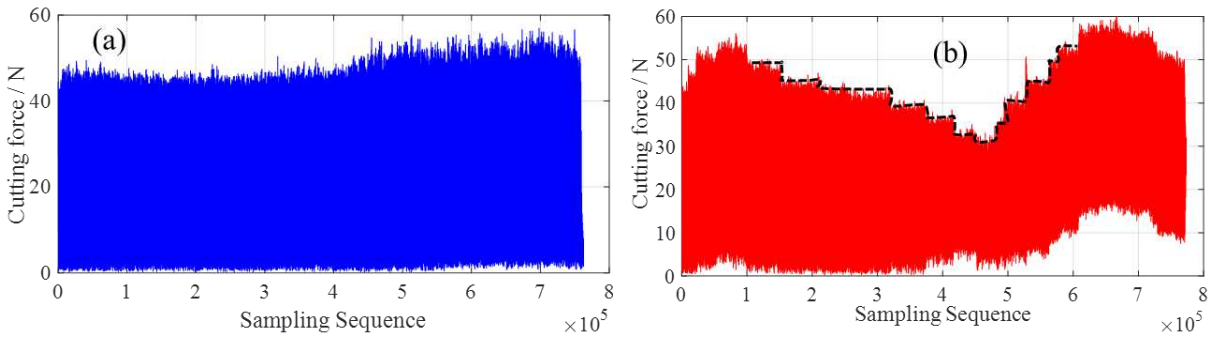

Fig11 The fluctuation of cutting force in zone 2. (a) The DHS algorithm. (b) The Mastercam method

As can be seen in Fig. 11, the cutting force of the DSH algorithm has no significant abrupt changes and the overall trend is relatively smooth as the curvature of the tool trajectory changes. In contrast, the cutting forces of the Mastercam method, in general, have significant abrupt changes and fluctuations. In addition, the black dashed line in the figure reflects the changing characteristics of the cutting forces, which means that the cutting forces between adjacent steps are also abruptly changing. Therefore, the DHS algorithm can make the machining process more stable compared to the Mastercam method.

6. Conclusion

(1) In this paper, we propose a least-squares-based toolpath smoothing and compression algorithm to address the problems of the huge amount of program code and low machining quality and efficiency in the traditional continuous short segment machining method. A simple and effective double-headed snake (DHS) algorithm is designed to implement the process by discretizing complex curves into a combination of straight and arc segments.

(2) Compared with the equal chord error method, D-P algorithm, and commercial CAM software, the DHS algorithm can obtain fewer trajectory data with the same constraint of contour error threshold, and the amount of trajectory data of the DHS algorithm is 63.93%, 74.4% and 75.56% less than the above three methods, respectively. In addition, when the number of discrete points of the trajectory is the same (e.g., n=100), the contour error of the DHS algorithm is reduced by more than one order of magnitude compared to the equal chord error method, D-P algorithm, and Mastercam. The maximum errors are reduced by 93.67%, 96.67%, and 97.29%, respectively.

The DHS algorithm is experimentally validated to be suitable for economical CNC machines and can achieve excellent machining quality. Therefore, in a practical sense, this method can enhance the ability of Low and mid-range machine tools to handle complex curves and further improve the quality and efficiency of machine tools for machining complex contour parts, thus improving the overall performance of these machines. Perhaps this will be a good solution for companies who are trying to improve the machining quality but lack high-end machine equipment. Therefore, in the subsequent research work, we intend to develop the corresponding function module in the sheet metal processing system (SheetCUT_CAM V1.0) owned by our laboratory. The G-code can be output by simply entering the spline curve, the accuracy threshold, and the tool information.

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Code availability: Not applicable.

Declarations:

Ethics approval and consent to participate: All authors agreed with the consent to participate.

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